The Analysis of Red Wine Quality Using Machine Learning model

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Introduction

In an attractive region of Portugal, the appeal of Vinho Verde which means "green wine" calls wine adventures and connoisseurs to discover the unique blend of taste and scenery (Winalist 2024) of its vineyards.

Wine quality is a critical factor in the wine industry, influencing consumer preferences, pricing, and market demand. The assessment of wine quality is traditionally performed by human experts known as connoisseurs. However, with the increasing availability of wine-related data, machine learning has emerged as a powerful tool (James et al. 2013; Weisberg 2014) that can predict wine quality based on its physicochemical properties.

This project aims to analyze the quality of red wine using machine learning techniques. By analyzing the various attributes of red wine, the objective is to identify the key factors that influence wine quality and develop predictive models to assess wine quality automatically. The insights gained from this study could help winemakers optimize their production processes, improve the overall product quality and reduce the request of connoisseurs to predict the quality of their wines.

Literature Review

The application of machine learning in enology (the study of wine) has progressed in recent years. Several studies have explored the use of various algorithms to predict wine quality based on physicochemical properties.

For instance, Cortez et al. (2009)(Cortez et al. 2009) used a dataset from the Vinho Verde region of Portugal to develop models using support vector machines (SVMs), neural networks, and random forests. Their study demonstrated that random forests and SVMs were particularly effective in predicting wine quality, with an accuracy exceeding 70%.

Other researchers have focused on feature selection to improve the predictive power of models. For example, González-Briones et al. (2018)(González-Briones et al. 2018) employed genetic algorithms to select the most relevant features from a wine dataset, resulting in improved model performance.

In this project, I build upon these existing studies by comparing the quality of red wine using machine learning, specifically linear regression and random forest. Random forests have been shown to be effective for high-dimensional data (Breiman 2001; Liu et al. 2018), while linear regression is often applied for interpretable models (Ioannidis 2008). I also explored the importance of different physicochemical properties in determining wine quality and assessed the models' performance across both types of wine.

Problem/Purpose

The goal of this project is to determine which physicochemical parameters significantly impact the quality of red wine, specifically focusing on the "Vinho Verde" variety produced in northern Portugal. By applying machine learning, the study seeks to predict which variables—such as volatile acidity, chlorides, and alcohol—are most associated with determining red wine quality. This approach builds on previous studies in enology that have used machine learning to model wine quality while focusing exclusively on red wine for a more targeted analysis.

Data Preprocessing

The dataset used in this study consists of physicochemical properties of 12 variables, along with their quality ratings on a scale from 0 to 10. The data preprocessing steps included:

- **Dependent variable**: Quality of the wine (on a scale from 1 (worst) to 10 (best)).
- **Independent variables**: Fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulfates, alcohol.

1. **Data Loading**: The dataset was loaded into R Markdown.
2. **Missing Values**: A check for missing values was performed, revealing no missing data.

Table of the independent and dependent variables

Variables	Regressors	Coefficients
y	quality	β_0
x1	fixed.acidity	β_1
x2	volatile.acidity	β_2
x3	citric.acid	β_3
x4	residual.sugar	β_4
x5	chlorides	β_5
x6	free.sulfur.dioxide	β_6
x7	total.sulfur.dioxide	β_7
x8	density	β_8
x9	pH	β_9
x10	sulphates	β_{10}
x11	alcohol	β_{11}

Prediction of the initial red wine quality based on all measured regressors

The equation for linear regression is:

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n$$

where y is the dependent variable (quality), $x_1, x_2, \ldots, x_n$ are the independent variables or regressors, and $\beta_0, \beta_1, \ldots, \beta_n$ are the coefficients.

Coefficient Table

Coefficients_red	Estimate	Std. Error	t value	Pr(>	t
(Intercept)	21.97	21.19	1.036	0.3002	
fixed.acidity	0.02499	0.02595	0.963	0.3357	
volatile.acidity	-1.084	0.1211	-8.948	< 2e-16	***
citric.acid	-0.1826	0.1472	-1.240	0.2150	
residual.sugar	0.01633	0.01500	1.089	0.2765	
chlorides	-1.874	0.4193	-4.470	8.37e-06	***
free.sulfur.dioxide	0.004361	0.002171	2.009	0.0447	*
total.sulfur.dioxide	-0.003265	0.0007287	-4.480	8.00e-06	***
density	-17.88	21.63	-0.827	0.4086	
pH	-0.4137	0.1916	-2.159	0.0310	*
sulphates	0.9163	0.1143	8.014	2.13e-15	***
alcohol	0.2762	0.02648	10.429	< 2e-16	***

Significance Code

Significance Code	Definition
***	0.001
**	0.01
*	0.05
.	0.1
	1

Regression Model Equation

For the red wine data, the linear regression model equation based on the regressors without considering the p-values is:

$$\text{quality} = 21.97 + (0.02499 \times \text{fixed.acidity}) - (1.084 \times \text{volatile.acidity}) - (0.1826 \times \text{citric.acid}) + (0.01633 \times \text{residual.sugar}) - (1.874 \times \text{c}$$

Significant Variables (p-value < 0.05)

The model with p-values less than the significance level ($\alpha = 0.05$) includes:

$$\text{quality} = -(1.084 \times \text{volatile.acidity}) - (1.874 \times \text{chlorides}) + (0.004361 \times \text{free.sulfur.dioxide}) - (0.003265 \times \text{total.sulfur.dioxide}) - (0.4$$

Analysis

There are several regressors that stand out as being significantly associated with red wine quality, since they have p-values less than the significance level ($\alpha = 0.05$). These regressors are:

Variable	Description
x2	volatile.acidity
x5	chlorides
x6	free.sulfur.dioxide
x7	total.sulfur.dioxide
x9	pH
x10	sulphates
x11	alcohol

Volatile acidity, chlorides, total sulfur dioxide, and pH have a significant negative effect on wine quality, meaning that higher levels of these components tend to decrease wine quality. On the other hand, alcohol and sulphates have strong positive effects, suggesting that higher concentrations of these components lead to better wine quality.

Model Performance

This model has an R-squared of 0.3606, suggesting that it explains about 36% of the fitted data in the regression model in the wine quality data.

Exploratory Data Analysis

Now, we will observe a plot of the residuals vs the predicted response check the
linearity assumption and **constant variance**

Normality of the red wine quality

Figure 1: Normality of the red wine quality

The plot looks satisfactory. The residuals does not show any obvious pattern. Hence, it has a constant variance and linearity assumption is satisfied.

Normality

Next, I had to observe **normality**

Histogram of the red wine quality

Figure 2: Histogram of the red wine quality

From my observation, the dependent/predictive variable, quality, looks like a normal distribution with a slight skewness to the right. The Q-Q (quantile-quantile) plot below also shows that most of the data are on line but the tail ends do show some skewness which proves the skewness on the histogram.

QQ plot of the red wine quality

Figure 3: QQ plot of the red wine quality

Shapiro_Wilk test

I performed the Shapiro-Wilk test for normality on the residuals of the model. The test statistic is given by:

$W = 0.99087$

The corresponding p-value is:

$$p\text{-value} = 1.954 \times 10^{-8}$$

Since the p-value is very small, I reject the null hypothesis, indicating that the residuals are not normally distributed.

Observation of the correlation matrix among all 11 regressors and the predictive variable (quality), in order to visually determine which variables are closely associated.

Correlation plot of the red wine quality

Figure 4: Correlation plot of the red wine quality

The regressors that have the most effect on quality are, volatile.acidity (-0.39)/alcohol (0.48), and sulphates (0.25). This observation can also be seen on the summary output of the m0_red model and that there is correlation between citric acid and fixed acidity. It also shows a significant correlation between free sulfur dioxide and total sulfur dixoide. Multicollinearity might be present between density and fixed.acidity, fixed.acidity and citric.acid.

Since **multicollinearity** was found in the correlation matrix, a Variance Inflation Factor(VIF) test was conducted to confirm our assumption that it does exist in the model.

Regressors	VIF
fixed.acidity	7.767512
volatile.acidity	1.789390
citric.acid	3.128022
residual.sugar	1.702588
chlorides	1.481932
free.sulfur.dioxide	1.963019
total.sulfur.dioxide	2.186813
density	6.343760
pH	3.329732
sulphates	1.429434
alcohol	3.031160

Regressors above may be causing multicollinearity in model 11 based on the VIF:

$fixedacidity = 7.767512$ $density = 6.343760$

It has been confirmed that there are normality and multicollinearity issues, the outliers has to be addressed to see if the coefficient of determination (R-squared) will improve.

By removing the outliers, we can see that R-squared has improved by only a small amount of 0.066 from 0.361 to 0.427.

I would compare both the data with and without outliers.

Best Subsets Regression With Outliers

Model Index and Predictors

Model Index	Predictors
1	alcohol
2	volatile..acidity alcohol
3	volatile..acidity sulphates alcohol
4	volatile..acidity total.sulfur.dioxide sulphates alcohol
5	volatile..acidity chlorides total.sulfur.dioxide sulphates alcohol
6	volatile..acidity chlorides total.sulfur.dioxide pH sulphates alcohol
7	volatile..acidity chlorides free.sulfur.dioxide total.sulfur.dioxide pH sulphates alcohol
8	volatile..acidity citric.acid chlorides free.sulfur.dioxide total.sulfur.dioxide pH sulphates alcohol
9	volatile..acidity citric.acid residual.sugar chlorides free.sulfur.dioxide total.sulfur.dioxide pH sulphates alcohol
10	fixed.acidity volatile..acidity citric.acid residual.sugar chlorides free.sulfur.dioxide total.sulfur.dioxide pH sulphates alcohol

Model Index	Predictors
11	fixed.acidity volatile..acidity citric.acid residual.sugar chlorides free.sulfur.dioxide total.sulfur.dioxide density pH sulphates alcohol

Subsets Regression Summary with Outliers

Model	R-Square	Adj. R-Square	Pred R-Square	C(p)	AIC	SBIC	SBC	MSEP	FPE	HSP	APC
1	0.2267	0.2263	0.2246	324.1115	3448.1135	1090.3748	3464.2449	806.8797	0.5052	3e-04	0.7752
2	0.3170	0.3161	0.3139	102.0818	3251.6275	1286.4844	3273.1360	713.1345	0.4468	3e-04	0.6856
3	0.3359	0.3346	0.3316	57.1879	3208.7683	1329.2376	3235.6540	693.8409	0.4350	3e-04	0.6674
4	0.3438	0.3421	0.3386	39.6184	3191.6693	1346.2787	3223.9321	686.0331	0.4304	3e-04	0.6603
5	0.3515	0.3495	0.3454	22.4791	3174.7667	1363.0770	3212.4066	678.3967	0.4259	3e-04	0.6534
6	0.3572	0.3548	0.3501	10.3837	3162.7015	1375.0322	3205.7185	672.8782	0.4227	3e-04	0.6485
7	0.3595	0.3567	0.3515	6.6823	3158.9769	1378.6948	3207.3711	670.8952	0.4217	3e-04	0.6470
8	0.3599	0.3567	0.3510	7.5516	3159.8391	1377.8080	3213.6105	670.8399	0.4219	3e-04	0.6473
9	0.3602	0.3565	0.3501	8.9404	3161.2237	1376.4025	3220.3722	671.0041	0.4223	3e-04	0.6479
10	0.3603	0.3562	0.3491	10.6832	3162.9648	1374.6439	3227.4904	671.3182	0.4227	3e-04	0.6486
11	0.3606	0.3561	0.3483	12.0000	3164.2766	1373.3075	3234.1793	671.4524	0.4231	3e-04	0.6491

Best Subsets Regression Without Outliers

Model Index and Predictors

Model Index	Predictors
1	alcohol
2	volatile..acidity alcohol
3	volatile..acidity sulphates alcohol
4	volatile..acidity total.sulfur.dioxide sulphates alcohol
5	volatile..acidity chlorides total.sulfur.dioxide sulphates alcohol
6	volatile..acidity chlorides total.sulfur.dioxide pH sulphates alcohol
7	volatile..acidity chlorides free.sulfur.dioxide total.sulfur.dioxide pH sulphates alcohol
8	volatile..acidity citric.acid chlorides free.sulfur.dioxide total.sulfur.dioxide pH sulphates alcohol
9	volatile..acidity citric.acid residual.sugar chlorides free.sulfur.dioxide total.sulfur.dioxide pH sulphates alcohol
10	fixed.acidity volatile..acidity citric.acid residual.sugar chlorides free.sulfur.dioxide total.sulfur.dioxide pH sulphates alcohol

Model Index	Predictors
11	fixed.acidity volatile..acidity citric.acid residual.sugar chlorides free.sulfur.dioxide total.sulfur.dioxide density pH sulphates alcohol

Subsets Regression Summary without Outliers

Model	R-Square	Adj. R-Square	Pred R-Square	C(p)	AIC	SBIC	SBC	MSEP	FPE	HSP	APC
1	0.2267	0.2263	0.2246	324.1115	3448.1135	1090.3748	3464.2449	806.8797	0.5052	3e-04	0.7752
2	0.3170	0.3161	0.3139	102.0818	3251.6275	1286.4844	3273.1360	713.1345	0.4468	3e-04	0.6856
3	0.3359	0.3346	0.3316	57.1879	3208.7683	1329.2376	3235.6540	693.8409	0.4350	3e-04	0.6674
4	0.3438	0.3421	0.3386	39.6184	3191.6693	1346.2787	3223.9321	686.0331	0.4304	3e-04	0.6603
5	0.3515	0.3495	0.3454	22.4791	3174.7667	1363.0770	3212.4066	678.3967	0.4259	3e-04	0.6534
6	0.3572	0.3548	0.3501	10.3837	3162.7015	1375.0322	3205.7185	672.8782	0.4227	3e-04	0.6485
7	0.3595	0.3567	0.3515	6.6823	3158.9769	1378.6948	3207.3711	670.8952	0.4217	3e-04	0.6470
8	0.3599	0.3567	0.3510	7.5516	3159.8391	1377.8080	3213.6105	670.8399	0.4219	3e-04	0.6473
9	0.3602	0.3565	0.3501	8.9404	3161.2237	1376.4025	3220.3722	671.0041	0.4223	3e-04	0.6479
10	0.3603	0.3562	0.3491	10.6832	3162.9648	1374.6439	3227.4904	671.3182	0.4227	3e-04	0.6486
11	0.3606	0.3561	0.3483	12.0000	3164.2766	1373.3075	3234.1793	671.4524	0.4231	3e-04	0.6491

AIC: Akaike Information Criteria
SBIC: Sawa's Bayesian Information Criteria
SBC: Schwarz Bayesian Criteria
MSEP: Estimated error of prediction, assuming multivariate normality
FPE: Final Prediction Error
HSP: Hocking's Sp
APC: Amemiya Prediction Criteria

I have decided not to remove the outliers because there are no significant difference in subset regression summary.

Proceeding to eliminate the regressors in search of finding the best model with all of the data.

From the above table, the models show that we have 11 models that stand out as being candidates for the **best model**, so our analysis will be on those 11 models to find the "best" model. So far it appears that model 7 and 8 may be the best fit as there AIC is lowest (3158.9769 and 3159.8391) and the highest Adjusted R-Square(0.3567). The analysis on all 11 models has to be conducted to confirm if my assumption is right.

Predictor	Model 1	Model 2	Model 3	Model 4	Model 5
volatile..acidity	1	1.0427	1.1111	1.1144	1.1349
chlorides					1.2800
free.sulfur.dioxide					
total.sulfur.dioxide				1.0515	1.0527
pH					
sulphates			1.0751	1.0812	1.3159
alcohol		1.0427	1.0446	1.0873	1.1622

Predictor	Model 6	Model 7	Model 8	Model 9	Model 10	Model 11
volatile..acidity	1.2280	1.2418	1.6415	1.6539	1.7599	1.7894
chlorides	1.3333	1.3333	1.3682	1.3703	1.4689	1.4819
free.sulfur.dioxide		1.8827	1.9198	1.9392	1.9487	1.9630
total.sulfur.dioxide	1.0538	1.9439	1.9933	2.0086	2.1732	2.1868
pH	1.2187	1.2546	1.6092	1.6102	2.2394	3.3297
sulphates	1.3212	1.3219	1.3294	1.3347	1.3415	1.4294
alcohol	1.2187	1.2202	1.2690	1.2796	1.2996	3.0312
citric.acid			2.1433	2.1836	3.1278	3.1280
residual.sugar				1.0935	1.0994	1.7026
density						6.3438
fixed.acidity					2.9755	7.7675

Cross-Validation Test

A **cross-validation test** was conducted to determine the best model by dividing the data into 5 folds for 5-fold cross-validation, which was repeated 10 times. Here, K = 5 represents 5-fold cross-validation, and R = 10 represents the repetitions of cross-validation. The Root Mean Squared Prediction Error (RTMSPE) is used to evaluate model performance.

Model	RTMSPE
LS1_red	0.4093523
LS2_red	0.3925461
LS3_red	0.3840450
LS4_red	0.3755269
LS5_red	0.3718223
LS6_red	0.3716930
LS7_red	0.3704254
LS8_red	0.3709433
LS9_red	0.3705801
LS10_red	0.3701532
LS11_red	0.3700179

LS11_red is the best-performing model, with the lowest Root Mean Squared Prediction Error (RTMSPE) of 0.3700179 among the evaluated models.

Model Training Results

After training my data, I evaluated each model based on RMSE (Root Mean Squared Error) and R-squared values. The lower the RMSE, the better the model fits the data. The highest R-squared value also indicates

a better fit, as it shows the proportion of variance explained by the model.

Model	RMSE	R-squared
Model 1	0.7154018	0.2259184
Model 2	0.6561272	0.3501944
Model 3	0.6442288	0.3740412
Model 4	0.6406302	0.3812956
Model 5	0.6377251	0.3865862
Model 6	0.6389678	0.3836629
Model 7	0.6360662	0.389635
Model 8	0.6354588	0.3909449
Model 9	0.6351809	0.391473
Model 10	0.6355821	0.3905961
Model 11	0.6351877	0.3913797

Model 9, with an RMSE of 0.6351809 and R-squared of 0.391473, is the best-performing model in terms of balancing predictive performance and model complexity. The inclusion of additional predictors beyond Model 9 did not significantly improve predictive performance, which suggests that adding more complexity could lead to overfitting.

Random Forest Algorithm Results

To further explore model performance, I applied the Random Forest machine learning algorithm to the data. The Random Forest results for each model are as follows:

Model	RMSE	R-squared
Model 1	0.7275283	0.2042848
Model 2	0.6582797	0.3461047
Model 3	0.6265142	0.4060919
Model 4	0.6001329	0.4585666
Model 5	0.5920438	0.4783019
Model 6	0.5869056	0.482932
Model 7	0.5815796	0.4948308
Model 8	0.5818991	0.4969148
Model 9	0.5805145	0.4981917
Model 10	0.5835927	0.4935075
Model 11	0.5829446	0.4953917

Based on the Random Forest results, Model 9 appears to be the most effective model with the lowest RMSE (0.5805145) and the highest R-squared (0.4981917). This indicates that Model 9 provides a good balance between model complexity and predictive accuracy. Adding more predictors beyond Model 9 did not significantly improve performance, suggesting that further complexity might lead to overfitting.

Conclusion

Model 9 consistently performed the best across both Linear Regression and Random Forest algorithms. Its predictive accuracy, with the lowest RMSE and highest R-squared, demonstrates that the inclusion of certain key physicochemical properties such as volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, pH, sulphates, and alcohol strikes the right balance between complexity and accuracy.

As noted by James et al. (2013) (James et al. 2013) and Murphy (2012) (Murphy 2012), complex models may perform well on training data but risk overfitting when more predictors are added without improving performance. Thus, Model 9 is the optimal model for predicting wine quality based on physicochemical properties.

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